

Theorem 2.1.1

$$1) A + B = B + A$$

Matrix addition is commutative so order of addition does not change the result

$$2) A + (B + C) = A + B + C$$

Associative, matrix additions can be grouped differently without affecting results

3) Existence of zero matrix O :

$$O + A = A \text{ for each } A$$

Each $m \times n$ matrix has a zero matrix, no changes if added

4) For each A exists an $m \times n$ matrix, $-A$, where $A + (-A) = O$

Negatives, additive inverses, which when added to A , brings A to O

$$5) K(A + B) = KA + KB$$

$$6) (K + p)A = KA + pA$$

Distribute, multiplying a matrix sum by a scalar is the same as scaling each matrix and then adding them

Distribute, multiplying a matrix by a scalar sum is the same as multiplying the matrix individually by each, and adding the results

$$7) (Kp)A = K(pA)$$

Associative, scalar multiplication order doesn't change the result

$$8) 1A = A$$

Multiplying by 1 doesn't change the matrix

Theorem 2.1.2

1) If A is an $n \times m$ matrix, A^T is an $m \times n$ matrix

The transpose of a matrix, obtained by swapping the rows and columns while maintaining order of appearance ($A_1 \rightarrow C_1$)

2) $(A^T)^T = A$ ← Transposing a transpose matrix returns the original matrix

3) $(kA)^T = kA^T$ ← A matrix A multiplied by a scalar k , when transposed, results in the same as multiplying the transpose by the scalar

4) $(A+B)^T = A^T + B^T$

↑
Transpose of a matrix sum is the same as the sum of their transposed forms

• T2.1.1 outlines the behaviour of matrix properties when added, multiplied by scalars.

T2.1.2 further describes these behaviours in conjunction with matrix transpose, a mechanism for representing A where $m \times n$ is instead $n \times m \rightarrow A^T$

N2.2.1

$$\begin{vmatrix} \cancel{0} & \cancel{5} & \cancel{-3} & \cancel{1} \\ \cancel{4} & \cancel{1} & \cancel{2} & \cancel{1} \\ \cancel{3} & \cancel{5} & \cancel{-5} & \cancel{-5} \end{vmatrix} + \begin{vmatrix} 1 & 6 & -7 & \cancel{-\frac{3}{2}} \\ \cancel{0} & \cancel{X} & \cancel{X} & \cancel{5} \\ \cancel{X} & \cancel{X} & \cancel{X} & \cancel{4} \\ \cancel{-2} & \cancel{-2} & \cancel{-6} & \cancel{-10} \\ \cancel{X} & \cancel{X} & \cancel{X} & \cancel{X} \end{vmatrix}$$

N2.2.2 a)

$$\begin{vmatrix} \cancel{2} & \cancel{5} \\ \cancel{0} & \cancel{-10} & \cancel{-25} \\ \cancel{4} & \cancel{5} \\ \cancel{20} & \cancel{-25} \\ \cancel{4} & \cancel{4} \\ \cancel{-20} & \cancel{-20} \end{vmatrix} + \begin{vmatrix} \cancel{0} & \cancel{5} \\ \cancel{0} & \cancel{-2} \\ \cancel{-1} & \cancel{2} \\ \cancel{2} & \cancel{4} \\ \cancel{2} & \cancel{5} \\ \cancel{-6} & \cancel{-10} \end{vmatrix} = \begin{vmatrix} -10 & -27 \\ 22 & -21 \\ -26 & -30 \end{vmatrix}$$

b) $A = \begin{vmatrix} -5 & -5 \\ 4 & 3 \\ 5 & -2 \end{vmatrix} \therefore A^T = \begin{vmatrix} -5 & 4 & -5 \\ -5 & 3 & -2 \end{vmatrix}$

Exercices 2.1

$$a) \begin{vmatrix} 3 & 2 & 1 \\ 5 & 1 & 0 \end{vmatrix} + \begin{vmatrix} 7 & 0 & -2 \\ -15 & 0 & 10 \\ 1 & -1 & 2 \end{vmatrix} = \begin{vmatrix} -12 & 2 & 11 \\ 0 & 6 & -10 \end{vmatrix}$$

-5 5 -10

$$b) 3 \begin{vmatrix} 3 \\ -1 \end{vmatrix} - 5 \begin{vmatrix} 6 \\ 2 \end{vmatrix} + 7 \begin{vmatrix} 1 \\ -1 \end{vmatrix}$$

$$= \begin{vmatrix} 9 \\ -3 \end{vmatrix} + \begin{vmatrix} -30 \\ -10 \end{vmatrix} + \begin{vmatrix} 7 \\ -7 \end{vmatrix} = \begin{vmatrix} -14 \\ -20 \end{vmatrix}$$

$$c) \begin{vmatrix} -2 & 11 \\ 3 & 2 \end{vmatrix} - 4 \begin{vmatrix} 1 & -2 \\ 0 & 1 \end{vmatrix} + 3 \begin{vmatrix} 2 & -2 \\ 1 & -2 \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix}$$

0 4 -3 -6

$$d) \begin{vmatrix} 3 & -1 & 2 \\ -2 & 9 & 3 \\ 1 & 1 & -6 \end{vmatrix} = \begin{vmatrix} -12 & 4 & -12 \end{vmatrix}$$

-18 -6 -8

$$e) \begin{vmatrix} 1 & -5 & 4 & 0 \\ 2 & 1 & 0 & 6 \end{vmatrix} = A \quad A^T = \begin{vmatrix} 1 & 2 \\ -5 & 1 \\ 4 & 0 \\ 0 & 6 \end{vmatrix}$$

$$g) \begin{vmatrix} 3 & -1 \\ 2 & 1 \end{vmatrix} - 2 \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix}^T$$

$$= \begin{vmatrix} 3 & -1 \\ 2 & 1 \end{vmatrix} + \begin{vmatrix} 1 & -2 \\ -2 & 1 \\ 4 & -2 \end{vmatrix} = \begin{vmatrix} 1 & -3 \\ 6 & -1 \end{vmatrix}$$

2.1.3 Exercises

$$A = \begin{matrix} 2 \times 2 \\ \begin{vmatrix} 2 & 1 \\ 0 & -1 \end{vmatrix} \end{matrix} \quad B = \begin{matrix} 2 \times 3 \\ \begin{vmatrix} 3 & -1 & 2 \\ 0 & 1 & 4 \end{vmatrix} \end{matrix} \quad C = \begin{matrix} 2 \times 2 \\ \begin{vmatrix} 3 & -1 \\ 2 & 0 \end{vmatrix} \end{matrix}$$

$$D = \begin{matrix} 3 \times 2 \\ \begin{vmatrix} 1 & 3 \\ -1 & 0 \\ 1 & 4 \end{vmatrix} \end{matrix} \quad E = \begin{matrix} 2 \times 3 \\ \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} \end{matrix}$$

a) $3A - 2B$ $A_{mn} = 2, 2$ but $B_{mn} = 2, 3$
 $\therefore$ incompatible for sum

$$b) 5C = 5 \begin{vmatrix} 3 & -1 \\ 2 & 0 \end{vmatrix} = \begin{vmatrix} 15 & -5 \\ 10 & 0 \end{vmatrix}$$

$$c) 3E^T = 3 \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix}^T = 3 \begin{vmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{vmatrix} = \begin{vmatrix} 3 & 0 \\ 0 & 3 \\ 3 & 0 \end{vmatrix}$$

d) $B + D$ $B_{mn} = 2, 3$ but $D_{mn} = 3, 2$
 $\therefore$ incompatible for sum

$$e) 4A^T - 3C = \begin{vmatrix} 2 & 0 \\ 1 & -1 \end{vmatrix}^T + \begin{vmatrix} -3 & -1 \\ 2 & 0 \end{vmatrix}^T = \begin{vmatrix} -1 & 3 \\ -2 & -4 \end{vmatrix}$$

f) $(A+C)^T$ Good example of Theorem 2.1.2 + addition without scalars 2.1.1

$$(A+C)^T = A^T + C^T$$

$$\rightarrow \begin{vmatrix} 2 & 0 \\ 1 & -1 \end{vmatrix} + \begin{vmatrix} 3 & 2 \\ -1 & 0 \end{vmatrix} = \begin{vmatrix} 5 & 2 \\ 0 & -1 \end{vmatrix}$$

$$\rightarrow \left(\begin{vmatrix} 2 & 1 \\ 0 & -1 \end{vmatrix} + \begin{vmatrix} 3 & -1 \\ 2 & 0 \end{vmatrix} \right)^T = \begin{vmatrix} 5 & 0 \\ 2 & -1 \end{vmatrix}^T = \begin{vmatrix} 5 & 2 \\ 0 & -1 \end{vmatrix}$$

Notice i, j, i, j persists across transpose

2.1.14 Exercises

A matrix is symmetric if $A = A^T$

$$a) \begin{vmatrix} 1 & s \\ -2 & t \end{vmatrix} = A, \quad A^T = \begin{vmatrix} 1 & -2 \\ s & t \end{vmatrix}$$

$$\begin{matrix} 1 & = & 1 \\ s & & -2 \\ -2 & = & s \\ t & = & t \end{matrix} \quad \left. \begin{matrix} \\ \\ \end{matrix} \right\} \begin{matrix} s = -2, \text{ so } A = A^T \\ \text{only when } s = -2 \\ \text{and } t \in \mathbb{R} \end{matrix}$$

$$b) \begin{vmatrix} s & t \\ st & 1 \end{vmatrix} = A, \quad A^T = \begin{vmatrix} s & st \\ t & 1 \end{vmatrix}$$

$$\begin{matrix} s & = & s \\ t & & st \\ st & = & t \\ 1 & = & 1 \end{matrix} \quad \left. \begin{matrix} \\ \\ \end{matrix} \right\} \begin{matrix} t - st = 0 \\ t(1-s) = 0 \end{matrix}$$

For this to be true, either $t=0$, $1-s=0$

$$t(1-s) = 0$$

$$t(0) = 0$$

$$1-s = 0 \Rightarrow s = 1$$

$\therefore A = A^T$ only when either $t=0$ or $s=1$, $s, t \in \mathbb{R}$

$$c) \begin{vmatrix} s & 2s & st \\ t & -1 & s \\ t & s^2 & s \end{vmatrix} = A$$

$$A^T = \begin{vmatrix} s & t & t \\ 2s & -1 & s^2 \\ st & s & s \end{vmatrix}$$

$$s=0, t=2s=0$$

$$\begin{vmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{vmatrix} = A \text{ so } A = A^T$$

$$s=1, t=2s=2$$

$$\begin{vmatrix} 1 & 2 & 2 \\ 2 & -1 & 1 \\ 2 & 1 & 1 \end{vmatrix} = A, \quad A^T = \begin{vmatrix} 1 & 2 & 2 \\ 2 & -1 & 1 \\ 2 & 1 & 1 \end{vmatrix} \quad \therefore A = A^T$$

$$A = A^T$$

$$\begin{matrix} s & = & s \\ 2s & = & t \\ st & = & t \\ t & = & -1 \\ s & = & s^2 \\ st & = & s \end{matrix} \quad \left. \begin{matrix} \\ \\ \end{matrix} \right\} \begin{matrix} t = 2s, \quad st = t \\ = s2s = t \\ = 2s^2 = t \\ = 2s^2 = t \\ = s^2 - s = 0 \\ = s(s-1) = 0 \end{matrix}$$

Only true when $s=0, s=1$

2.1.19 Exercises

$$a) (A + \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \end{pmatrix})^T = \begin{pmatrix} 2 & 1 \\ 0 & 5 \\ 3 & 8 \end{pmatrix} \quad \begin{matrix} \leftarrow \\ \end{matrix} \begin{matrix} 4 \\ \end{matrix} B = A + \begin{pmatrix} 3 & -3 & 0 \\ 3 & 6 & 12 \end{pmatrix}$$

$$B^T = \begin{pmatrix} 2 & 1 \\ 0 & 5 \\ 3 & 8 \end{pmatrix}$$

$$A = B - \begin{pmatrix} 3 & -3 & 0 \\ 3 & 6 & 12 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 0 & 3 \\ 1 & 5 & 2 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & 0 & 3 \\ 1 & 5 & 2 \end{pmatrix} - \begin{pmatrix} 3 & -3 & 0 \\ 3 & 6 & 12 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 3 & 3 \\ -2 & -1 & -4 \end{pmatrix}$$

$$b) (3A^T + 2 \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix})^T = \begin{pmatrix} 8 & 0 \\ 3 & 1 \end{pmatrix} \quad \begin{matrix} D \\ B \end{matrix} \quad \begin{matrix} \begin{pmatrix} 8 & 3 \\ 0 & 1 \end{pmatrix} \\ B^T \end{matrix}$$

$$(3A^T + 2D)^T = B$$

$$= 3A + 2D^T = B$$

$$= 3A^T + 2D = B^T \quad \begin{pmatrix} 8 & 3 \\ 0 & 1 \end{pmatrix} - 2D \quad \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$$

$$= 3A^T = \begin{pmatrix} 6 & 3 \\ 0 & -3 \end{pmatrix} / 3$$

$$= A^T = \begin{pmatrix} 2 & 1 \\ 0 & -1 \end{pmatrix}, \quad A = \begin{pmatrix} 2 & 0 \\ 1 & -1 \end{pmatrix}$$

$$d) (2A^T - 5 \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix})^T = 4A - 9 \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} \rightarrow (2A^T - 5D)^T$$

$$E = 4A - 9E$$

$$2A - 5D^T = 4A - 9E + 5D^T$$

$$-4A$$

$$2A = 5D^T - 9E \Rightarrow 2A = 9E - 5D^T \quad (\text{cont.})$$

2.1.15 Exercise d continued

$$2A = 9E - 5D^T \quad 9E = \begin{vmatrix} 9 & 9 \\ -9 & 0 \end{vmatrix}$$

$$D^T = \begin{vmatrix} 1 & -1 \\ 0 & 2 \end{vmatrix} \quad 5D^T = \begin{vmatrix} 5 & -5 \\ 0 & 10 \end{vmatrix}$$

$$2A = \begin{vmatrix} 9 & 9 \\ -9 & 0 \end{vmatrix} - \begin{vmatrix} 5 & -5 \\ 0 & 10 \end{vmatrix} = \begin{vmatrix} 4 & 14 \\ -9 & -10 \end{vmatrix}$$

$$A = \begin{vmatrix} 2 & 7 \\ -\frac{9}{2} & -5 \end{vmatrix}$$

Theorem 2.2.3

Assuming x_1 is any solution to a system $Ax = b$, every solution x_2 to $Ax = b$ can be expressed

$x_2 = x_0 + x_1$ where x_0 is a solution to the homogeneous system $Ax = 0$

In other words, any specific solution for $Ax = b$ can be added with a solution of $Ax = 0$ to get any solution for the original system

Theorem 2.2.4

For $Ax = b$ system with aug. $|A/b|$

$|A/b|$ rank is either r , $r+1$

$r = \text{rank of } A \text{ system}$

$Ax = b$ only consistent when $\text{rank } |A/b| = r$

only inconsistent when $\text{rank } |A/b| = r+1$

NOTE: $Ax = b$ is compact notation,

$|A/b|$ combines coeffs and constants for matrix representation

Theorem 2.2.6 Good verification tool to check A and B are the same transformation
If $Ax = Bx$ for $x \in \mathbb{R}$, $A = B$

Summary, 2.2.3: How to find more solutions with an existing solution by using the associated $Ax = 0$

2.2.4 outlines whether solution exists based on rank

Definition 2.8

T_A represents the transformation T brought on by the matrix A , e.g. take matrix A , T_A describes the transformation applied when a vector (example) is exposed to A arithmetic

Ex 2.2.13

A transformation $\mathbb{R}^2$ carries vector $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ to its reflection $\begin{bmatrix} a_1 \\ -a_2 \end{bmatrix}$, where the y value is inverted to create a reflection along the x axis. This can be carried out alternatively by multiplying $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, so for every vector x in $\mathbb{R}^2$, each x is carried to the vector Ax in $\mathbb{R}^2$ where $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow f: \mathbb{R} \rightarrow \mathbb{R}$, a function carrying x to $f(x)$

Ex 2.2.15

Let $R_{\frac{\pi}{2}}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ representing a counterclockwise rotation from the origin by $\frac{\pi}{2}$ radians (90°)
 $R_{\frac{\pi}{2}}$ is induced by $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $x = \begin{bmatrix} a \\ b \end{bmatrix}$, $R_{\frac{\pi}{2}}(x) = \begin{bmatrix} -b \\ a \end{bmatrix}$
 when we multiply $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0a - 1b \\ 1a + 0b \end{bmatrix} = \begin{bmatrix} -b \\ a \end{bmatrix}$

Ex 2.2.16

when $a > 0$, transformation $T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax \\ y \end{bmatrix}$ induced by $A = \begin{bmatrix} a & 0 \\ 0 & 1 \end{bmatrix}$ is 1) an x -expansion of $\mathbb{R}^2$ when $a > 1$
 2) an x -compression when $0 < a < 1$ (cont.)

Ex 2.2-16 (continued)

Expansion: $a > 1$, x coordinates are scaled by 'a'

compression: $0 < a < 1$, x coordinates are scaled by a value less than 1 but greater than 0, shrinking x coords

If $b > 0$ and $A = \begin{bmatrix} 1 & 0 \\ 0 & b \end{bmatrix}$, we get y -expansions and compressions. To compare:

$a > 0$ for $A = \begin{bmatrix} a & 0 \\ 0 & 1 \end{bmatrix}$

a is a scalar targeting an x coord while persisting y coords

$b > 0$ for $A = \begin{bmatrix} 1 & 0 \\ 0 & b \end{bmatrix}$

b is a scalar targeting a y coord while persisting x coords

N2.S.2

Let $A = \begin{pmatrix} -2 & -1 \\ -2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $u = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ $v = \begin{pmatrix} 3 \\ -3 \end{pmatrix}$

a) $Au =$

$Av =$

$Bu =$

$Bv =$

$-2(-1) + -1(0)$
 $-2(-1) + 1(0)$

$-2(3) + -1(-3)$
 $-2(3) + 1(-3)$

incompatible incompatible

$= \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

$= \begin{bmatrix} -3 \\ -9 \end{bmatrix}$

b) The transformation A is $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ when

$n =$

and $m =$

$A = \begin{bmatrix} -2 & -1 \\ -2 & 1 \end{bmatrix}$ is a 2×2 matrix, $m=2$, $n=2$

$\therefore T_A: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

c) For general vector $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$, determine $T_A(x)$:

$Ax = \begin{bmatrix} -2 & -1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2x_1 + -1x_2 \\ -2x_1 + 1x_2 \end{bmatrix} = \begin{bmatrix} -2x_1 - x_2 \\ -2x_1 + x_2 \end{bmatrix}$

$$\begin{aligned}
 2) \ a) \quad & x_1 - x_2 + 3x_3 = 5 \\
 & -3x_1 + x_2 + x_3 = -6 \\
 & 5x_1 - 8x_2 = 9
 \end{aligned}$$

$$\begin{vmatrix} 1 & -1 & 3 \\ -3 & 1 & 1 \\ 5 & -8 & 0 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \\ x_3 \end{vmatrix} = \begin{vmatrix} 5 \\ -6 \\ 9 \end{vmatrix} \Rightarrow x_1 \begin{vmatrix} 1 \\ -3 \\ 5 \end{vmatrix} + x_2 \begin{vmatrix} -1 \\ 1 \\ -8 \end{vmatrix} + x_3 \begin{vmatrix} 3 \\ 1 \\ 0 \end{vmatrix} = \begin{vmatrix} 5 \\ -6 \\ 9 \end{vmatrix}$$

= RHS

↑ Very important, the system is expressed by $\begin{vmatrix} \text{coeff} \end{vmatrix} \times \begin{vmatrix} \text{vars} \end{vmatrix}$, the other way around $\neq Ax=b$

$$\therefore x_1 \begin{vmatrix} 1 \\ -3 \\ 5 \end{vmatrix} + x_2 \begin{vmatrix} -1 \\ 1 \\ -8 \end{vmatrix} + x_3 \begin{vmatrix} 3 \\ 1 \\ 0 \end{vmatrix} = \begin{vmatrix} 5 \\ -6 \\ 9 \end{vmatrix} \quad (ii)$$

3) Drawing on Definition 2.5 (i), Theorem 2.25
Solve Ax for each

$$a) \ A = \begin{vmatrix} 3 & -2 & 0 \\ 5 & -4 & 1 \end{vmatrix}, \ x = \begin{vmatrix} x_1 \\ x_2 \\ x_3 \end{vmatrix}$$

$$i) = a_1 \begin{vmatrix} 3 \\ 5 \end{vmatrix}, a_2 \begin{vmatrix} -2 \\ -4 \end{vmatrix}, a_3 \begin{vmatrix} 0 \\ 1 \end{vmatrix}$$

$$\begin{aligned}
 Ax &= x_1 a_1 + x_2 a_2 + x_3 a_3 \\
 &= \begin{vmatrix} 3x_1 - 2x_2 + 0x_3 \\ 5x_1 - 4x_2 + x_3 \end{vmatrix} = \begin{vmatrix} 3x_1 - 2x_2 \\ 5x_1 - 4x_2 + x_3 \end{vmatrix}
 \end{aligned}$$

↑ 2×1 vector

$$ii) \ R_1 = \begin{vmatrix} 3 & -2 & 0 \end{vmatrix} \quad R_2 = \begin{vmatrix} 5 & -4 & 1 \end{vmatrix}$$

$$3x_1 - 2x_2 + 0x_3 = \begin{vmatrix} 3x_1 - 2x_2 \end{vmatrix}$$

$$5x_1 - 4x_2 + x_3 = \begin{vmatrix} 5x_1 - 4x_2 + x_3 \end{vmatrix}$$

Exercise 2.2. (continued)

$$\begin{aligned} 5) d) \quad & 2x_1 + x_2 - x_3 - x_4 = -1 \\ & 3x_1 + x_2 + x_3 - 2x_4 = -2 \\ & -x_1 = x_2 + 2x_3 + x_4 = 2 \\ & -2x_1 - x_2 + 2x_4 = 3 \end{aligned}$$

$$\begin{array}{cccc|c}
 2 & 1 & -1 & -1 & 1 \\
 1 & 1 & 2 & 1 & 2 \\
 3 & 1 & 1 & -2 & -2 \\
 \hline
 -1 & -1 & 2 & 1 & 2 \\
 2 & 1 & -1 & -1 & 1 \\
 -2 & 1 & 0 & 2 & 3 \\
 \hline
 1 & 0 & 1 & 0 & 1 \\
 1 & 1 & 2 & 1 & 2 \\
 0 & 1 & -3 & -1 & -3 \\
 \hline
 0 & 2 & 7 & 1 & 7 \\
 0 & 0 & 1 & -1 & -2 \\
 0 & 1 & -4 & 0 & -1 \\
 0 & 0 & -1 & 1 & 2
 \end{array}$$

$$\begin{array}{l}
 R1 \leftrightarrow R3 \\
 \xrightarrow{x-1} \begin{array}{cccc|c} 1 & 1 & -2 & -1 & 2 \end{array} \\
 \xrightarrow{-2R1} \begin{array}{cccc|c} 0 & -1 & 3 & 3 & -2 \end{array} \xrightarrow{x-1} \begin{array}{cccc|c} 0 & 1 & -3 & -1 & -2 \end{array} \\
 \xrightarrow{-5R1} \begin{array}{cccc|c} 0 & -2 & 7 & 1 & 4 \end{array} \\
 \xrightarrow{+2R1} \begin{array}{cccc|c} 0 & 1 & -4 & 0 & -1 \end{array} \\
 \xrightarrow{-R3} \begin{array}{cccc|c} 1 & 0 & 0 & 1 & 3 \end{array} \\
 \xrightarrow{+3R3} \begin{array}{cccc|c} 0 & 1 & -3 & -1 & -3 \end{array} \xrightarrow{+7R3} \begin{array}{cccc|c} 0 & 1 & 0 & -4 & -9 \end{array} \\
 \xrightarrow{+2R2} \begin{array}{cccc|c} 0 & 2 & 7 & 1 & 7 \end{array} \xrightarrow{-R1} \begin{array}{cccc|c} 0 & 0 & 1 & -1 & -2 \end{array} \\
 \xrightarrow{+R3} \begin{array}{cccc|c} 0 & 0 & -1 & 1 & 2 \end{array} \xrightarrow{+R3} \begin{array}{cccc|c} 0 & 0 & 0 & 0 & 0 \end{array}
 \end{array}$$

$$\begin{aligned} x_1 &= 3 - x_4 & \text{e.g. } x_4 = 0 &= \begin{vmatrix} 3 \\ -9 \\ -2 \\ 0 \end{vmatrix} & A(0) &= \begin{vmatrix} -1 \\ 4 \\ 1 \\ 1 \end{vmatrix} \\ x_2 &= -9 + 4x_4 \\ x_3 &= -2 + x_4 \\ x_4 &= t & \begin{vmatrix} 3 \\ -9 \\ -2 \\ 0 \end{vmatrix} + t \begin{vmatrix} -1 \\ 4 \\ 1 \\ 1 \end{vmatrix} & , t \in \mathbb{R} \end{aligned}$$

Theorem 2.3.3

1) $IA = A$ and $AI = A$ where I is an identity matrix e.g.

$$\begin{array}{ccc} 1 \times 1 & 2 \times 2 & 3 \times 3 \\ I & \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \end{array}$$

Multiplying matrix A by identity matrix always results in the original matrix A .

- 2) Associativity, $A(BC) = (AB)C$, grouping matrices does not change the outcome
- 3) Distribution, $A(B+C) = AB+AC$, matrix mult distribute over matrix addition left-to-right.
- 4) Commutativity, $(B+C)A = BA+CA$, distributing A over the matrix sum $B+C$ is the same as multiplying each B, C by A and adding them
- 5) $a(AB) = (aA)B = A(aB)$ where a is a scalar, a can multiply the whole matrix product, or multiply each matrix before the product
- 6) $(AB)^T = B^T A^T$, transposing the product of two matrices is the same as their transpose product if done in reverse order.

Note for 4: This commutativity works for matrices despite matrix multiplication being generally non-commutative

N2.4.2 4 $A = \begin{bmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix}$ $B = \begin{bmatrix} 6 & -2 & 4 \\ 0 & 1 & 3 \\ 7 & 7 & 5 \end{bmatrix}$

$A^2 = A \times A$

$$\begin{bmatrix} -3 & 12 & 76 \\ 48 & 29 & 98 \\ 24 & 56 & 97 \end{bmatrix}$$

$(3 \times 3) + (-2 \times 6) + (7 \times 0) = -6$
 $(3 \times -2) + (-2 \times 5) + (7 \times 4) = 28$
 $(3 \times 7) + (-2 \times 4) + (7 \times 9) = 63$
 $(6 \times 3) + (5 \times 6) + (4 \times 0) = 30$
 $(6 \times -2) + (5 \times 5) + (4 \times 4) = 16$
 $(6 \times 7) + (5 \times 4) + (4 \times 9) = 36$
 $(0 \times 3) + (4 \times 6) + (9 \times 0) = 24$
 $(0 \times -2) + (4 \times 5) + (9 \times 4) = 20$
 $(0 \times 7) + (4 \times 4) + (9 \times 9) = 81$

$A^2_{a_1} = 3x_1 + 6x_2 + 0x_3$

Note: a linear combo is expressed as a sum

$B^2 = B \times B$

$$\begin{bmatrix} 64 & 14 & 38 \\ 21 & 22 & 18 \\ 77 & 28 & 74 \end{bmatrix}$$

$(6 \times 6) + (-2 \times 0) + (4 \times 7) = 28$
 $(6 \times -2) + (-2 \times 1) + (4 \times 3) = 20$
 $(6 \times 4) + (-2 \times 3) + (4 \times 5) = 24$
 $(0 \times 6) + (1 \times 0) + (3 \times 7) = 21$
 $(0 \times -2) + (1 \times 1) + (3 \times 3) = 10$
 $(0 \times 4) + (1 \times 3) + (3 \times 5) = 18$
 $(7 \times 6) + (7 \times 0) + (5 \times 7) = 35$
 $(7 \times -2) + (7 \times 1) + (5 \times 3) = 25$
 $(7 \times 4) + (7 \times 3) + (5 \times 5) = 25$

Exercise 2.4.1

Find AB if $A = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 1 \end{pmatrix}$ $B = \begin{pmatrix} 3 & 1 & 5 & -1 \\ 4 & -2 & 1 & 0 \end{pmatrix}$

$$\begin{array}{l}
 \begin{array}{l}
 \overset{6}{(2 \times 3)} + \overset{4}{(1 \times 4)} \\
 \overset{2}{(2 \times 1)} + \overset{-2}{(1 \times -2)} \\
 \overset{3}{(2 \times 5)} + \overset{1}{(1 \times 1)} \\
 \overset{-2}{(2 \times -1)} + \overset{0}{(1 \times 0)} \\
 \overset{-1}{(2 \times 3)} + \overset{0}{(1 \times 4)} \\
 \overset{-1}{(2 \times 1)} + \overset{-2}{(1 \times -2)} \\
 \overset{-1}{(2 \times 5)} + \overset{0}{(1 \times 1)}
 \end{array}
 \end{array}
 \begin{array}{l}
 \begin{array}{l}
 \overset{10}{(3 \times 1)} + \overset{11}{(1 \times 1)} \\
 \overset{-3}{(3 \times -1)} + \overset{-5}{(1 \times -2)} \\
 \overset{13}{(3 \times 5)} + \overset{16}{(1 \times 1)} \\
 \overset{-1}{(3 \times -1)} + \overset{0}{(1 \times 0)} \\
 \overset{-3}{(3 \times 3)} + \overset{4}{(1 \times 4)} \\
 \overset{3}{(3 \times 1)} + \overset{-2}{(1 \times -2)} \\
 \overset{3}{(3 \times 5)} + \overset{-2}{(1 \times 1)} \\
 \overset{-3}{(3 \times -1)} + \overset{0}{(1 \times 0)}
 \end{array}
 \end{array}$$

Exercise 2.4.2

Find AB , BA if $A = \begin{pmatrix} 3 & 1 \\ 0 & 4 \end{pmatrix}$ $B = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$

$$\begin{array}{l}
 AB = \begin{array}{l}
 \overset{3}{(3 \times 1)} + \overset{0}{(1 \times 0)} \\
 \overset{3}{(3 \times 1)} + \overset{2}{(1 \times 2)} \\
 \overset{0}{(0 \times 1)} + \overset{4}{(4 \times 0)} \\
 \overset{0}{(0 \times 1)} + \overset{8}{(4 \times 2)}
 \end{array} = \begin{pmatrix} 3 & 5 \\ 0 & 8 \end{pmatrix}
 \end{array}$$

Another example of commutativity despite being matrix multiplication

$$\begin{array}{l}
 BA = \begin{array}{l}
 \overset{3}{(1 \times 3)} + \overset{0}{(1 \times 0)} \\
 \overset{1}{(1 \times 1)} + \overset{4}{(1 \times 4)} \\
 \overset{0}{(0 \times 1)} + \overset{0}{(2 \times 0)} \\
 \overset{0}{(0 \times 1)} + \overset{8}{(2 \times 4)}
 \end{array} = \begin{pmatrix} 3 & 5 \\ 0 & 8 \end{pmatrix}
 \end{array}$$

Exercise 2.4.5 (continued) $A = \begin{vmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{vmatrix}$

$I = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$ $5I = \begin{vmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{vmatrix}$

$(A-5I)(A+I) = 0$ $A \neq 5I$ & $A \neq -I$

$A-5I = \begin{vmatrix} -4 & 2 & 2 \\ 2 & -4 & 2 \\ 2 & 2 & -4 \end{vmatrix}$ $A+I = \begin{vmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{vmatrix}$

B

C

$BC = \begin{vmatrix} -4 & 2 & 2 \\ 2 & -4 & 2 \\ 2 & 2 & -4 \end{vmatrix} \begin{vmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{vmatrix} = \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix}$

$R_1: \begin{vmatrix} -4 & 2 & 2 \\ -8 & 4 & 4 \end{vmatrix}$
 $R_2: \begin{vmatrix} -4 & 2 & 2 \\ -8 & 4 & 4 \end{vmatrix}$
 $R_3: \begin{vmatrix} -4 & 2 & 2 \\ -8 & 4 & 4 \end{vmatrix}$

R_2, R_3 are both variants of R_1 , equal,

Exercise 2.4.6

$A = \begin{vmatrix} 3 & 2 & 7 \\ 1 & 0 & 4 \end{vmatrix}$ i) $AN^T = \begin{vmatrix} 62 & 25 \\ 25 & 17 \end{vmatrix}$

$N^T = \begin{vmatrix} 3 & -1 \\ 2 & 0 \\ 7 & 4 \end{vmatrix}$

$\begin{vmatrix} 3 & 2 & 7 \\ 1 & 0 & 4 \end{vmatrix} \begin{vmatrix} 3 & -1 \\ 2 & 0 \\ 7 & 4 \end{vmatrix} = \begin{vmatrix} 3 \times 3 + 2 \times 2 + 7 \times 7 \\ 3 \times (-1) + 2 \times 0 + 7 \times 4 \\ 1 \times 3 + 0 \times 2 + 4 \times 7 \end{vmatrix}$

$\begin{vmatrix} 3 \times 3 + 2 \times 2 + 7 \times 7 \\ 3 \times (-1) + 2 \times 0 + 7 \times 4 \\ 1 \times 3 + 0 \times 2 + 4 \times 7 \end{vmatrix} = \begin{vmatrix} 62 & 25 \\ 25 & 17 \end{vmatrix}$

$\begin{vmatrix} 3 \times 3 + 2 \times 2 + 7 \times 7 \\ 3 \times (-1) + 2 \times 0 + 7 \times 4 \\ 1 \times 3 + 0 \times 2 + 4 \times 7 \end{vmatrix} = \begin{vmatrix} 62 & 25 \\ 25 & 17 \end{vmatrix}$

ii) $A^T A = \begin{vmatrix} 10 & 6 & 17 \\ 6 & 4 & 14 \\ 17 & 14 & 65 \end{vmatrix}$

$\begin{vmatrix} 3 & 2 & 7 \\ 1 & 0 & 4 \end{vmatrix} \begin{vmatrix} 3 & -1 \\ 2 & 0 \\ 7 & 4 \end{vmatrix} = \begin{vmatrix} 3 \times 3 + 2 \times 2 + 7 \times 7 \\ 3 \times (-1) + 2 \times 0 + 7 \times 4 \\ 1 \times 3 + 0 \times 2 + 4 \times 7 \end{vmatrix}$